

SYLLABUS :-

Name of the subject Structural Analysis 2. L-T-P 3-1-0 3. Credits 4 4. Name of the Department Civil Engineering 5. Status of the subject a. Specify the Session, Semester: 2021-22 onwards .b. Please specify the level of the subject: B.Tech/DD .c. Whether the subject will be offered as core or elective: Core .d. The semester in which the subject will be offered: Spring .e. Name of the programme in whose curricula this subject will be included: Civil Engineering B.Tech .H. 4 years, Civil Engineering Dual-Degree .All. 6. Prerequisites for the subject, if any Solid Mechanics .CE21005. 7. Objectives This course emphasizes the classical approaches to compute the deflections and the internal forces in determinate and indeterminate structures. Both displacement and force methods of analyses are discussed. 8. Contents .Minimum 36 hours and preferably in multiple of 6 hours. .a. 39 to 42 hours for 3-0- 0 or 3-1-0 .b. 52 to 56 hours for 4-0- 0 Introduction to determinate and indeterminate structures, deflection of beams using direct integration method, moment curvature relations, moment-area theorems, and conjugate beam method, principle of virtual work and unit load method, force method for beams and trusses, displacement method for indeterminate beams and frames, influence line diagrams, cables and arches 9. Names of the faculty members of the Department/ Centers/ School who have the necessary expertise and will be the willing to teach the subject .Minimum two faculty members should be willing to teach the subject: Prof. L. S. Ramachandra, Prof. N. Dhang, Prof. B. Bhattacharya, Prof. D. Maity, Prof. A. Deb, Prof. S. Chakraborty, Prof. A. Shaw, Prof. B. Banerjee, Prof. S. J. Hossain, Prof. P. Patra, Prof. A. Chatterjee 10. Do the contents of the subject have an overlap with any other subject offered in the Institute. 11. Recommended text books and reference books 1. Structural Analysis by R.C. Hibbeler Text book 2. Structural Analysis by Devdas Menon ,Reference book.